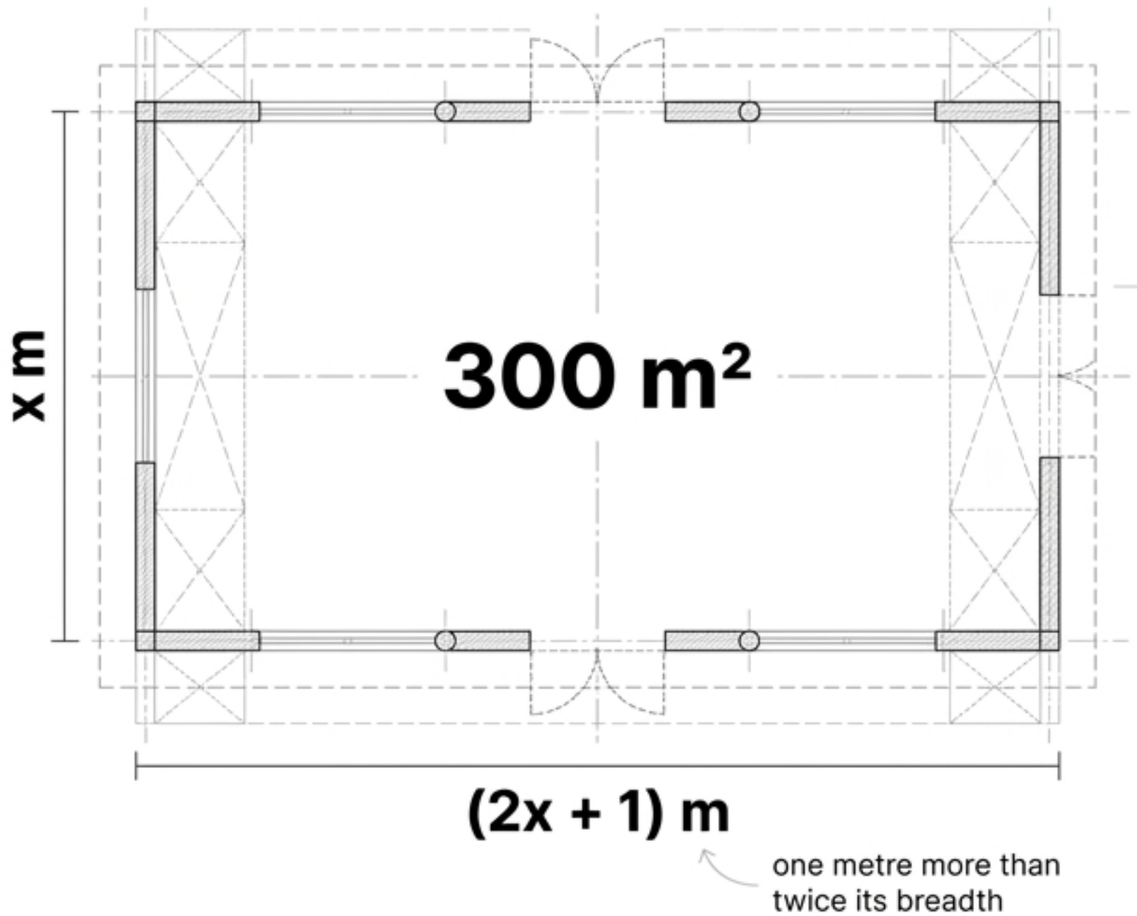
The Architect's Blueprint: Designing the Prayer Hall



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$$l \times b = A$$

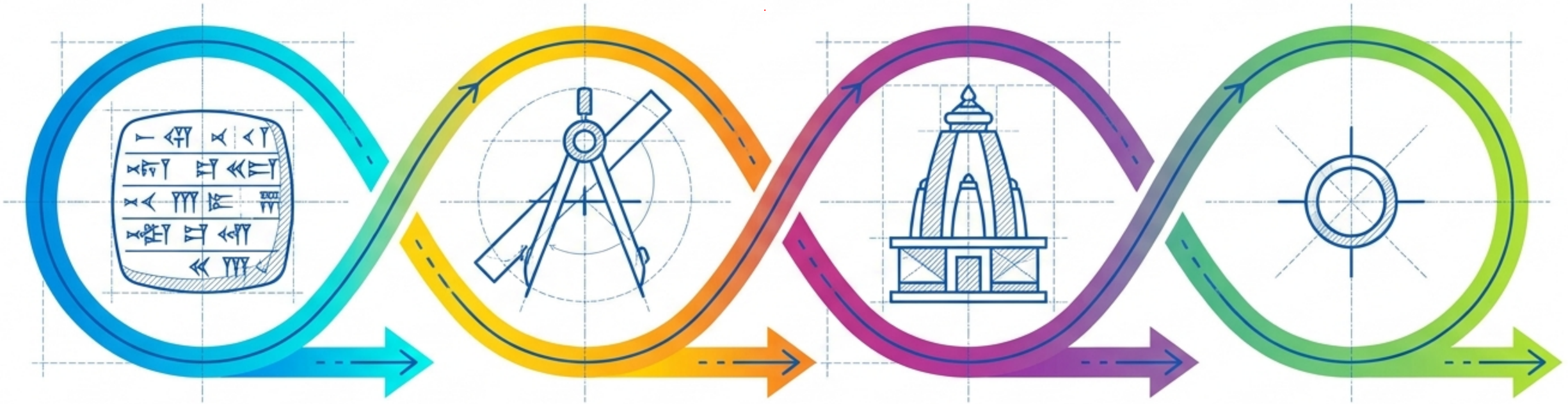
$$(2x+1)(x) = 300$$

$$\rightarrow 2x^2 + x = 300$$

$$2x^2 + x - 300 = 0$$

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AN ANCIENT INHERITANCE



Babylonians

The first to find two positive numbers with a given sum and product (solving $x^2 - px + q = 0$).

Euclid (Greece)

Developed a geometrical approach to finding lengths.

Brahmagupta (India, C.E. 598–665)

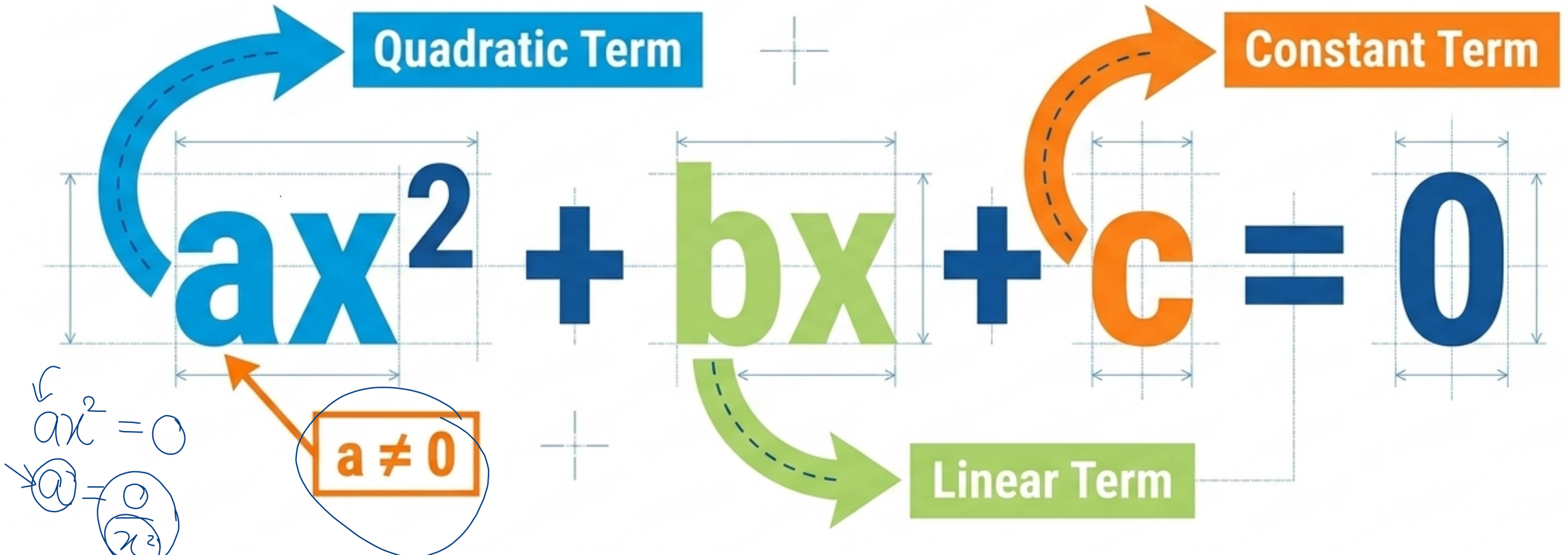
Gave an explicit formula to solve $ax^2 + bx = c$.

Sridharacharya (C.E. 1025)

Derived the formula using the method of completing the square.

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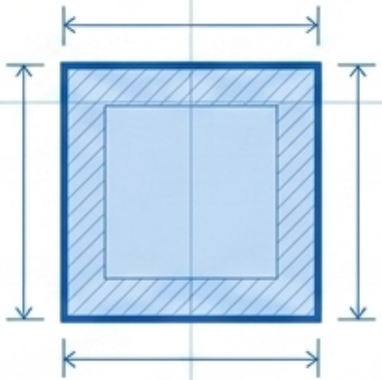
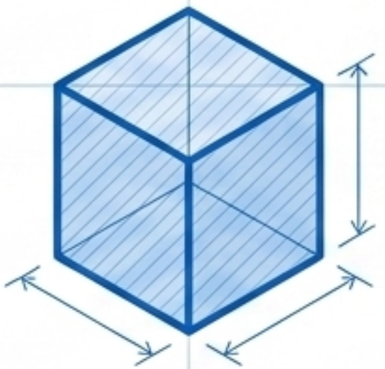
ANATOMY OF THE STANDARD FORM



A quadratic equation in the variable x is an equation of the form $ax^2 + bx + c = 0$, where a, b, c are real numbers.

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THE POLYNOMIAL LINE-UP

Linear	Quadratic	Cubic
$ax + b = 0$	$ax^2 + bx + c = 0$	$ax^3 + bx^2 + cx + d = 0$
		
1D Length	2D Area	3D Volume
Highest power is 1	Highest power is 2	Highest power is 3

NotebookLM

Example 1: Translating the Real World



John and Jivanti have 45 marbles. Both Both lost 5. Product of remaining is 124.

[Ans: $x^2 - 45x + 324 = 0$]



Production cost of a toy is 55 minus toys produced. Total cost = ₹750.

[Ans: $x^2 - 55x + 750 = 0$]

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Example 2: Disguised Quadratics

$(x - 2)^2 + 1 = 2x - 3$

$x^2 + 4 - 4x + 1 = 2x - 3$

$x^2 - 4x + 5 = 2x - 3$

$\Rightarrow x^2 - 6x + 8 = 0$

[Ans: $x^2 - 6x + 8 = 0$ (Yes)]

$x(x + 1) + 8 = (x + 2)(x - 2)$

$\Rightarrow x^2 + x + 8 = x^2 - 2x + 2x - 4$

$x^2 - x^2 + x + 8 + 4 = 0$

$x + 12 = 0$

$(a+b)^3 = a^3 + b^3 + 3ab(a+b)$

[Ans: $x + 12 = 0$ (No)]

$x(2x + 3) = x^2 + 1$

$2x^2 + 3x = x^2 + 1$

$2x^2 - x^2 + 3x - 1 = 0$

$x^2 + 3x - 1 = 0$

[Ans: $x^2 + 3x - 1 = 0$ (Yes)]

$(x + 2)^3 = x^3 - 4$

$x^3 + 2^3 + 3(x)(2)(x+2) = x^3 - 4$

$x^3 + 8 + 6x(x+2) = x^3 - 4$

$x^3 + 8 + 6x^2 + 12x = x^3 - 4$

$6x^2 + 12x + 12 = 0$

[Ans: $6x^2 + 12x + 12 = 0$ (Yes)]

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Exercise 4.1 (Q1, Parts i - iv): Check the Equations

(i) $(x + 1)^2 = 2(x - 3)$

[Ans: $x^2 + 7 = 0$ (Yes)]

(ii) $x^2 - 2x = (-2)(3 - x)$

[Ans: $x^2 - 4x + 6 = 0$ (Yes)]

(iii) $(x - 2)(x + 1) = (x - 1)(x + 3)$

[Ans: $3x - 1 = 0$ (No)]

(iv) $(x - 3)(2x + 1) = x(x + 5)$

[Ans: $x^2 - 10x - 3 = 0$ (Yes)]

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Exercise 4.1 (Q1, Parts v - viii): Check the Equations

$2x^2 - 6x - x + 3 = x^2 - x + 5x - 5$

$2x^2 - 7x + 3 = x^2 + 4x - 5$

(v) $(2x - 1)(x - 3) = (x + 5)(x - 1)$

$2x^2 - x^2 - 7x - 4x + 3 + 5 = 0$

$x^2 - 11x + 8 = 0$

[Ans: $x^2 - 11x + 8 = 0$ (Yes)]

$x^2 + 3x + 1 = x^2 - 4x + 4$

$\Rightarrow 3x + 4x + 1 - 4 = 0$

$7x - 3 = 0$

(vi) $x^2 + 3x + 1 = (x - 2)^2$

$(a-b)^2 = a^2 - 2ab + b^2$

[Ans: $7x - 3 = 0$ (No)]

$x^3 + 2^3 + 3(x)(2)(x+2) = 2x^3 - 2x$

$\Rightarrow x^3 + 8 + 6x^2 + 12x = 2x^3 - 2x$

$\Rightarrow x^3 + 6x^2 + 12x + 8 = 2x^3 - 2x$

(vii) $(x + 2)^3 = 2x(x^2 - 1)$

$\Rightarrow x^3 - 2x^3 + 6x^2 + 12x + 2x + 8 = 0$

$-x^3 + 6x^2 + 14x + 8 = 0$

[Ans: $x^3 - 6x^2 - 14x - 8 = 0$ (No)]

$x^3 - 4x^2 - x + 1 = x^3 - 2^3 - 3(x)(2)(x-2)$

$= x^3 - 8 - 6x(x-2)$

$x^3 - 4x^2 - x + 1 = x^3 - 8 - 6x^2 + 12x$

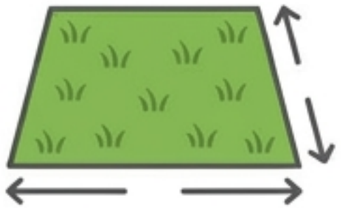
$\Rightarrow 2x^2 - 13x + 9 = 0$

(viii) $x^3 - 4x^2 - x + 1 = (x - 2)^3$

[Ans: $2x^2 - 13x + 9 = 0$ (Yes)]

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Exercise 4.1 (Q2): Mathematical Translation



(i) Area of rectangular plot is 528 m². Length is one more than twice its breadth.

[Ans: $2x^2 + x - 528 = 0$]

Let the breadth of rectangle is x m.

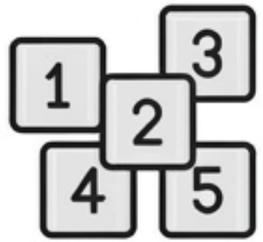
length will be $2x+1$.

Area of rectangle $= l \times b$

$528 = (2x+1)x$

$528 = 2x^2 + x$

$2x^2 + x - 528 = 0$



(ii) The product of two consecutive positive integers is 306.

[Ans: $x^2 + x - 306 = 0$]

Let the first positive int. is x

$x+1$

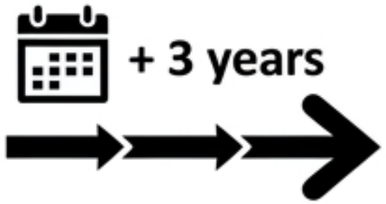
$x(x+1) = 306$

$x^2 + x = 306$

$x^2 + x - 306 = 0$

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Exercise 4.1 (Q2): Mathematical Translation



(iii) Rohan's mother is 26 years older than him. The product of their ages 3 years from now will be 360.

Let Rohan's present age is x.

Her mother's present age will be $26+x$

After 3 years

Rohan's age $= x+3$

Mother's age $= x+29$

$(x+3)(x+29) = 360$

$x^2+29x+3x+87 = 360$

$\boxed{x^2+32x-273 = 0}$

[Ans: $x^2 + 32x - 273 = 0$]



(iv) A train travels 480 km. If speed was 8 km/h less, it would take 3 hours more.

Let the required speed is x km/h

time taken will be $\frac{480}{x}$

Speed $= x-8$ km/h

time $= \boxed{\frac{480}{x}+3}$

$(x-8)\left(\frac{480}{x}+3\right) = 480$

$(x-8)\left(\frac{480+3x}{x}\right) = 480$

$(x-8)(480+3x) = 480x$

[Ans: $x^2 - 8x - 1280 = 0$]

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03/07/26

Solving Quadratic Equations

The Mechanism: Splitting the Middle Term

$2x^2-5x+3$

1. Original Equation $2x^2 - 5x + 3 = 0$

-3×-2

$0 \times 0 = 6$

$0+0=-5$

2. The Goal Find factors of $(2 \times 3) = 6$ that add up to -5.

3. The Split $-5x$ → $-2x$ and $-3x$ (i.e. $2x^2 - 2x - 3x + 3$)

4. Grouping $2x(x-1) - 3(x-1) = 0$

5. Linear Factors $(2x-3)(x-1) = 0$

Either $2x-3=0$, $2x=3$, $x=\frac{3}{2}$

OR $x-1=0$, $x=1$

The **zeroes** of the quadratic polynomial $ax^2 + bx + c$ and the roots of the quadratic equation $ax^2 + bx + c = 0$ are the **same.**

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Example 3 & 4: Finding Roots

Example 3

$$2x^2 - 5x + 3 = 0$$

[Ans: x = 1, x = 3/2]

Example 4

$$6x^2 - x - 2 = 0$$

$6x^2 - 4x + 3x - 2 = 0$

$2x(3x-2) + 1(3x-2) = 0$

$(3x-2)(2x+1) = 0$

Either $3x - 2 = 0$

$3x = 2$

$x = \frac{2}{3}$

or $2x + 1 = 0$

$2x = -1$

$x = \frac{-1}{2}$

[Ans: x = 2/3, x = -1/2]

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Example 5: Repeated Roots

$0 \times 0 = 6$

$0 + 0 = -2\sqrt{6}$

Find the roots of the quadratic equation:

$$3x^2 - 2\sqrt{6}x + 2 = 0$$

What happens when the factors are identical?

$3x^2 - \sqrt{6}x - \sqrt{6}x + 2 = 0$

$\sqrt{3}x(\sqrt{3}x - \sqrt{2}) - \sqrt{2}(\sqrt{3}x - \sqrt{2}) = 0$

$(\sqrt{3}x - \sqrt{2})(\sqrt{3}x - \sqrt{2}) = 0$

Either $\sqrt{3}x - \sqrt{2} = 0$ or $\sqrt{3}x - \sqrt{2} = 0$

$x = \sqrt{\frac{2}{3}}$ $x = \sqrt{\frac{2}{3}}$

[Ans: x = √(2/3), x = √(2/3)]

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Exercise 4.2 (Q1, Parts i - iii): Factorisation

(i) $x^2 - 3x - 10 = 0$

(ii) $2x^2 + x - 6 = 0$

$\sqrt{2}x^2 + 5x + 2x + 5\sqrt{2} = 0$

$x(\sqrt{2}x + 5) + \sqrt{2}(\sqrt{2}x + 5) = 0$

$(\sqrt{2}x + 5)(x + \sqrt{2})$

$x = -\frac{5}{\sqrt{2}}$ $\quad$ $x = -\sqrt{2}$

[Ans: x = -2, 3/2]

(iii) $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$

$\square \times \square = 10$

$\square + \square = 7$

[Ans: x = $-\sqrt{2}$, $-5/\sqrt{2}$]

Exercise 4.2 (Q1, Parts iv - v): Factorisation

(iv) $2x^2 - x + \frac{1}{8} = 0$

Multiply both sides of the eq by 8.

$16x^2 - 8x + 1 = 0$

$16x^2 - 4x - 4x + 1 = 0$

$4x(4x-1) - 1(4x-1) = 0$

$(4x-1)(4x-1) = 0$

$x = \frac{1}{4}$ and $\frac{1}{4}$

[Ans: x = 1/4, 1/4]

(v) $100x^2 - 20x + 1 = 0$

[Ans: x = 1/10, 1/10]

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Exercise 4.2 (Q2): Solving Example 1



Solve $x^2 - 45x + 324 = 0$
(John & Jivanti's Marbles)

[Ans: x = 36 or 9]



Solve $x^2 - 55x + 750 = 0$
(Cottage Industry Toys)

[Ans: x = 25 or 30]

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Exercise 4.2 (Q3 & Q4): Number Puzzles



2	182
7	91
13	13
	1

Q3. Find two numbers whose sum is 27 and product is 182.

Let the first number is x.

Second number will be $27-x$.

So $x(27-x)=182$

$27x-x^2=182$

$x^2-27x+182=0$

$x^2-13x-14x+182=0$

$x(x-13)-14(x-13)=0$

$x=13$ or 14

$-x^2+27x-182=0$

[Ans: 13, 14]



Q4. Find two consecutive positive integers, sum of whose squares is 365.

Let the two no's are x, $x+1$

$x^2+(x+1)^2=365$

$x^2+x^2+1+2x=365$

$2x^2+2x-364=0$

$2(x^2+x-182)=0$

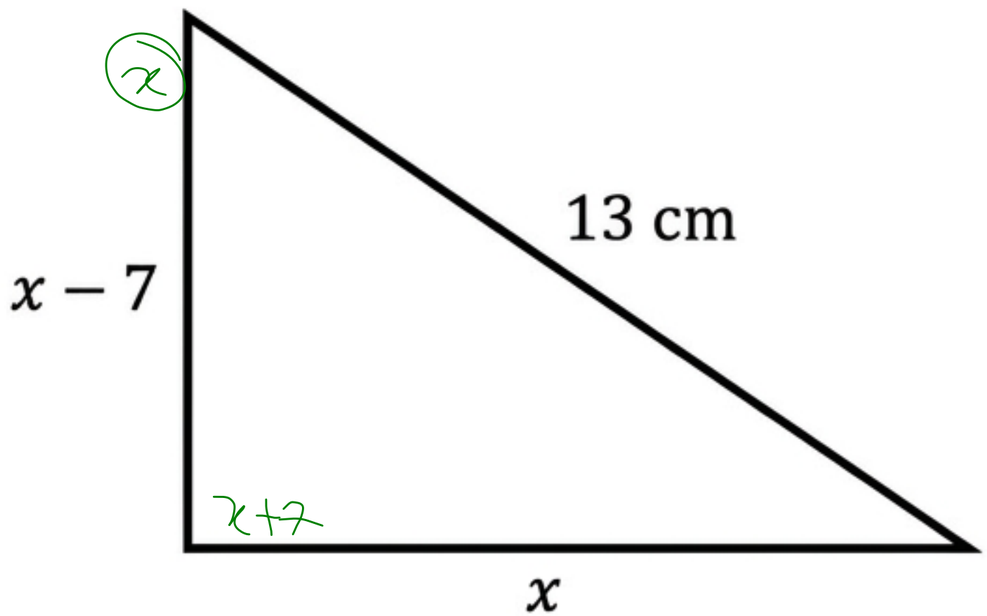
$x=13$ or -14

[Ans: 13, 14]

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Exercise 4.2 (Q5): The Right Triangle

Q5. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.



[Ans: base = 12 cm, altitude = 5 cm]

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Exercise 4.2 (Q6): The Pottery Workshop

Q6. A cottage industry produces pottery articles. Cost of production of each article was 3 more than twice the number of articles produced that day. Total cost of production on that day was ₹90. Find number of articles and cost of each.



[Ans: articles = 6, cost = ₹15]

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The Quadratic Playbook: Summary



The World — Real-life situations (Areas, Objects, Speeds)

translates to



The Language — Standard Form $ax^2 + bx + c = 0$ $(a \neq 0)$

solved via



The Mechanism — Splitting the middle term (Factorisation)

reveals



The Roots — The values of x where $a\alpha^2 + b\alpha + c = 0$.

What happens when you cannot split the middle term?
The **Quadratic Formula** awaits...

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